

Lahore Garrison University

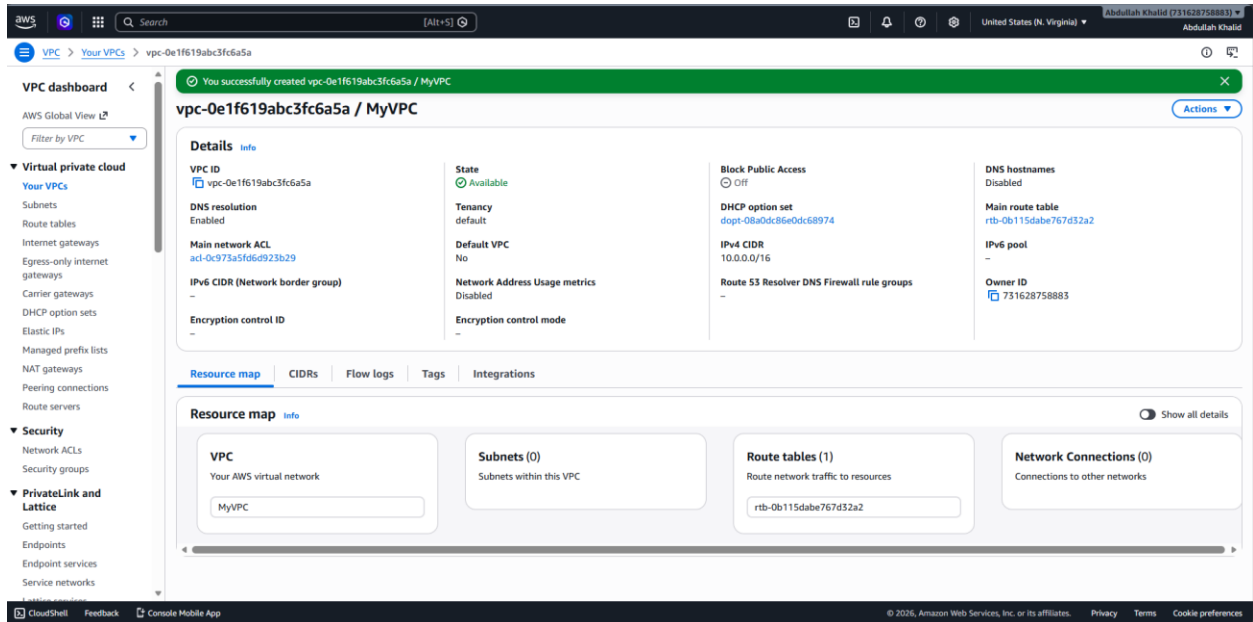
DHA Phase 6 Lahore



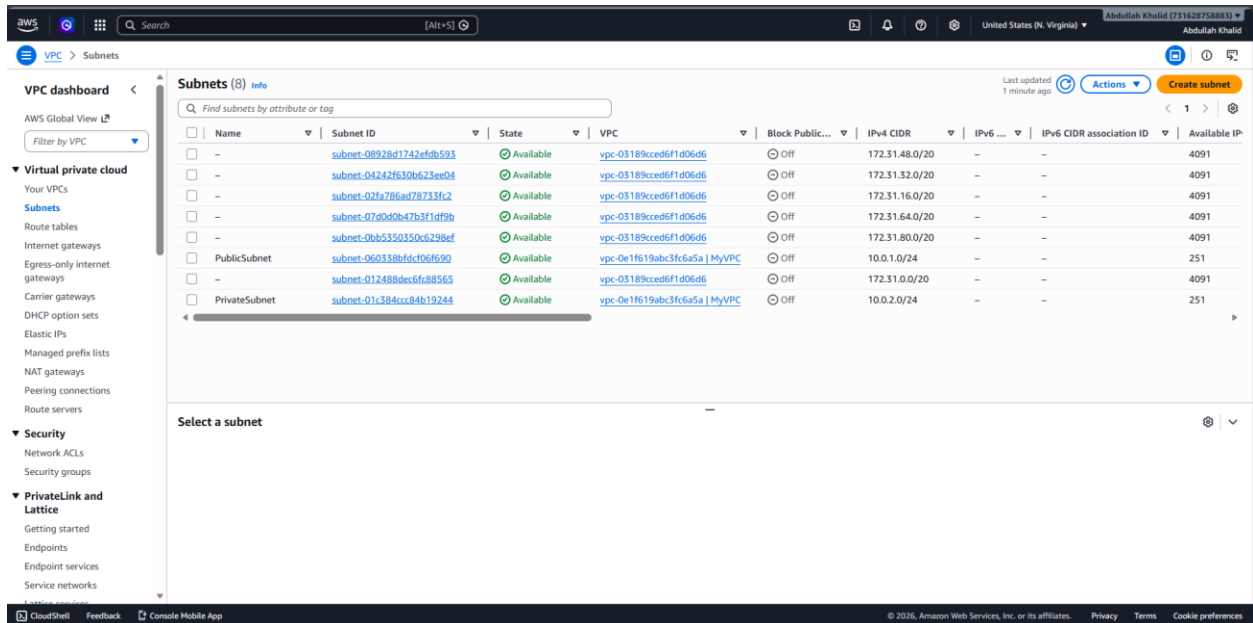
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Lab 9

Step 1: Created a Custom VPC



Step 2: Created public and Private Subnets



Step 3: Created and attached Internet Gateways to MyVPC

The screenshot shows the AWS Management Console interface for the 'Internet gateways' section. The left sidebar contains navigation links for VPC dashboard, Virtual private cloud, Security, and PrivateLink and Lattice. The main content area displays the details for the Internet gateway 'igw-0c31d36e8dcb80d5a / MyIGW'. The details section includes the Internet gateway ID, State (Attached), VPC ID, and Owner. Below this, there is a 'Tags' section showing one tag with the key 'Name' and value 'MyIGW'. A notification banner at the top indicates that the Internet gateway was successfully attached to the VPC.

Key	Value
Name	MyIGW

Step 4: Route Tables

The screenshot shows the AWS Management Console interface for the 'Route tables' section. The left sidebar contains navigation links for VPC dashboard, Virtual private cloud, Security, and PrivateLink and Lattice. The main content area displays the details for the route table 'rtb-0b115dabe767d32a2 / PublicRouteTable'. The details section includes the Route table ID, Main status (Yes), VPC, and Owner ID. Below this, there is a 'Routes' section showing two routes. The first route has a destination of 0.0.0.0/0 and a target of 'igw-0c31d36e8dcb...', with a status of 'Active'. The second route has a destination of 10.0.0.0/16 and a target of 'local', with a status of 'Active'.

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0c31d36e8dcb...	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Step 5: Instances created and tested over networks

Public:

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-0-1-232 ~]$ ping google.com
PING google.com (172.253.63.102): 56(84) bytes of data.
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=1 ttl=102 time=2.09 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=2 ttl=102 time=1.72 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=3 ttl=102 time=2.12 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=4 ttl=102 time=1.70 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=5 ttl=102 time=1.71 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=6 ttl=102 time=1.71 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=7 ttl=102 time=1.73 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=8 ttl=102 time=1.69 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=9 ttl=102 time=1.69 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=10 ttl=102 time=1.70 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=11 ttl=102 time=1.71 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=12 ttl=102 time=1.69 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=13 ttl=102 time=2.14 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=14 ttl=102 time=2.12 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=15 ttl=102 time=1.68 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=16 ttl=106 time=1.72 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=17 ttl=106 time=1.70 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=18 ttl=106 time=2.15 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=19 ttl=106 time=2.14 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=20 ttl=106 time=1.73 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=21 ttl=106 time=1.76 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=22 ttl=106 time=1.72 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=23 ttl=106 time=2.15 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=24 ttl=106 time=2.14 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=25 ttl=106 time=2.12 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=26 ttl=106 time=1.72 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=27 ttl=106 time=1.75 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=28 ttl=106 time=1.73 ms
64 bytes from bi-in-f102.1a100.net (172.253.63.102): icmp_seq=29 ttl=106 time=2.16 ms

i-0528ed2c5446c2a0b (PublicServer)
PublicPc: 44.214.141.216 PrivatePc: 10.0.1.232
```

Private:

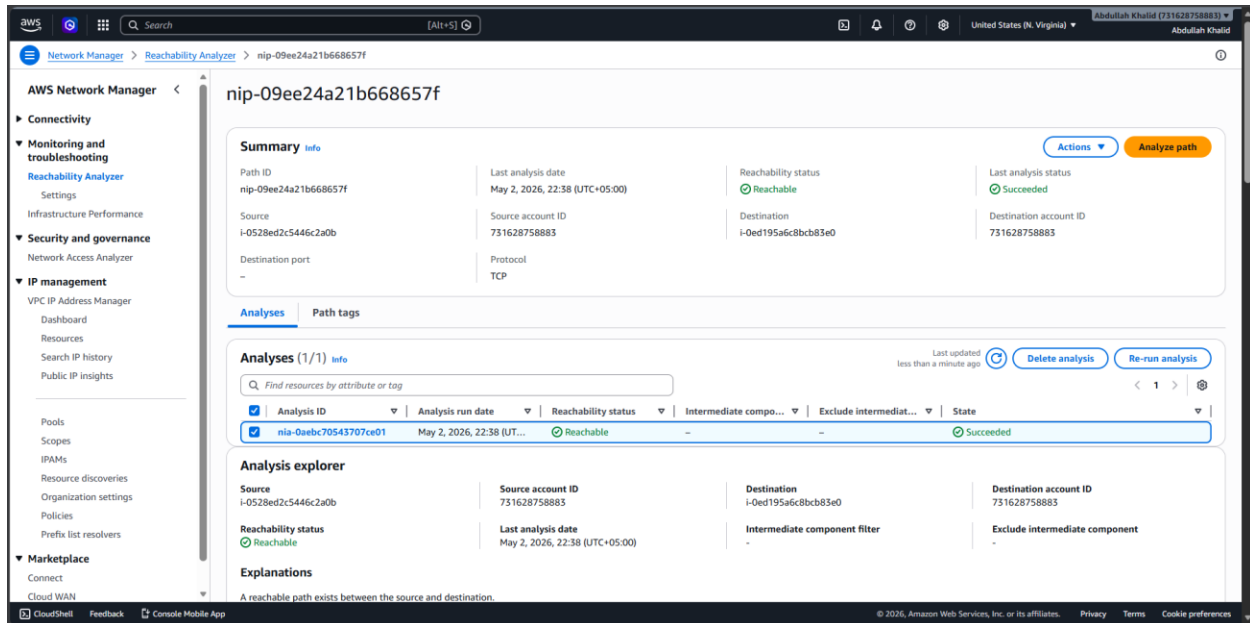
```
inst login: Sat May 2 15:44:54 2026 from 18.206.107.29
[ec2-user@ip-10-0-1-232 ~]$ ping 10.0.2.186
PING 10.0.2.186 (10.0.2.186): 56(84) bytes of data.
^C
--- 10.0.2.186 ping statistics ---
70 packets transmitted, 0 received, 100% packet loss, time 71754ms

[ec2-user@ip-10-0-1-232 ~]$ ping google.com
PING google.com (142.251.163.138): 56(84) bytes of data.
64 bytes from ww-in-f138.1a100.net (142.251.163.138): icmp_seq=1 ttl=105 time=1.25 ms
64 bytes from ww-in-f138.1a100.net (142.251.163.138): icmp_seq=2 ttl=105 time=1.28 ms
64 bytes from ww-in-f138.1a100.net (142.251.163.138): icmp_seq=3 ttl=105 time=1.27 ms
64 bytes from ww-in-f138.1a100.net (142.251.163.138): icmp_seq=4 ttl=105 time=1.22 ms
^C
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.218/1.256/1.283/0.024 ms

[ec2-user@ip-10-0-1-232 ~]$ ping 10.0.2.186
PING 10.0.2.186 (10.0.2.186): 56(84) bytes of data.
64 bytes from 10.0.2.186: icmp_seq=507 ttl=127 time=0.248 ms
64 bytes from 10.0.2.186: icmp_seq=508 ttl=127 time=0.244 ms
64 bytes from 10.0.2.186: icmp_seq=509 ttl=127 time=0.244 ms
64 bytes from 10.0.2.186: icmp_seq=510 ttl=127 time=0.244 ms
64 bytes from 10.0.2.186: icmp_seq=511 ttl=127 time=0.430 ms
64 bytes from 10.0.2.186: icmp_seq=512 ttl=127 time=0.248 ms
64 bytes from 10.0.2.186: icmp_seq=513 ttl=127 time=0.209 ms
64 bytes from 10.0.2.186: icmp_seq=514 ttl=127 time=0.236 ms
^2
[1]+ Stopped ping 10.0.2.186
[ec2-user@ip-10-0-1-232 ~]$ ping 10.0.2.186
PING 10.0.2.186 (10.0.2.186): 56(84) bytes of data.
64 bytes from 10.0.2.186: icmp_seq=1 ttl=127 time=0.307 ms
64 bytes from 10.0.2.186: icmp_seq=2 ttl=127 time=0.229 ms
64 bytes from 10.0.2.186: icmp_seq=3 ttl=127 time=0.229 ms
64 bytes from 10.0.2.186: icmp_seq=4 ttl=127 time=0.231 ms
64 bytes from 10.0.2.186: icmp_seq=5 ttl=127 time=0.236 ms
64 bytes from 10.0.2.186: icmp_seq=6 ttl=127 time=0.250 ms
^2
[2]+ Stopped ping 10.0.2.186

[ec2-user@ip-10-0-1-232 ~]$ ping 10.0.2.186
```

Step 6: VPC peering configured and communication verified



What I have done in this lab so far:

1. Created a **custom VPC** named MyVPC with CIDR 10.0.0.0/16.
2. Created a **public subnet** in MyVPC.
3. Created a **private subnet** in MyVPC.
4. Created and attached an **Internet Gateway** to MyVPC.
5. Configured the **public route table** with a 0.0.0.0/0 route to the Internet Gateway.
6. Associated the **public subnet** with the public route table.
7. Launched a **PublicServer EC2** instance in the public subnet.
8. Launched a **PrivateServer EC2** instance in the private subnet.
9. Verified that the **public instance** has **internet** access by **pinging google.com**.
10. Updated the **private instance** Security Group to allow SSH from the public instance.
11. Created a second VPC, **MyVPC2**, with CIDR 10.1.0.0/16.
12. Created a **subnet** inside MyVPC2.
13. Created a **VPC peering connection** between MyVPC and MyVPC2.

14. Added **peering routes** in the **route tables** so both VPCs know how to reach each other.

15. Opened **Reachability Analyzer** and started the **connectivity test** between your instances.

16. Prepared the **source** and **destination** for the final connectivity check.

So overall, I have worked on **VPC creation, subnets, route tables, internet gateway, EC2 instances, security groups, VPC peering, and reachability testing.**
